

Team ID = ECF5 - 169

PRODUCT & ENGINEERING OVERVIEW

NovaShields Plus

Smart Two-Wheeler Safety Ecosystem

1. Accident Detection + Tiered Escalation
2. Insurance Telemetry
3. Guardian Dashboard
4. Rider Telemetrics

Team Code Catalyst · Panipat Institute of Engineering & Technology (PIET)



THE PROBLEM

India's Two-Wheeler Crisis

~1.7L

Two Wheelers Road Fatalities

2024, MoRTH — the latest official count

46.2%

Fatalities of Total Fatalities Are Two-Wheeler Riders

The single largest share of any vehicle class

~20

Lives Lost Per Hour

70.8% in rural areas — EMS response is dangerously slow

⚠ Today's alerting: a single SMS event — no escalation, no family visibility, no evidence.

Why Existing Solutions Fail ?

Smart Helmets

· Expensive ₹8k–20k
· Rider-dependent
· Poor adoption

Smartphone Apps

· Fail when phone is damaged or battery dies
· No dedicated sensors

Premium Connected Bikes

· Only on ₹1.5L+ motorcycles
· Excludes 95% of riders

Standard Motorcycles

· Zero crash detection
· No ride data
· No emergency dispatch

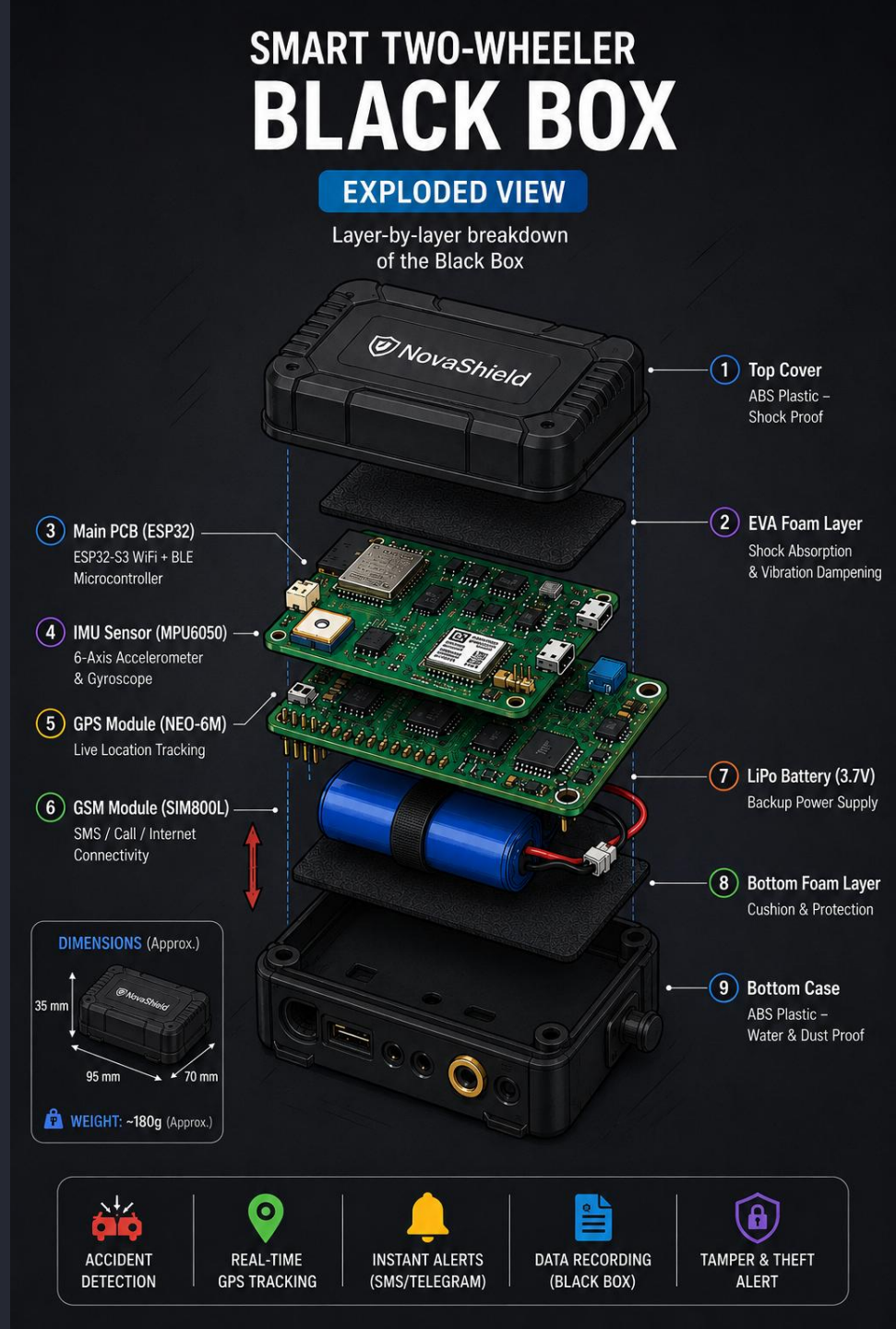
"The gap is not another app. The gap is an always-on safety layer for every two-wheeler."

OUR APPROACH

Four Pillars. One Kit.

No new sensors required.

1	**Tiered Emergency Escalation** Right responder at the right time, every time.
2	**Insurance Telemetry** Pre-crash black box, UBI scoring, fraud evidence.
3	**Guardian Dashboard** Consent-based family visibility, geofence alerts.
4	**Trip & Journey Report** Per-ride metrics, safety trends, gamified streaks.



System Architecture & Hardware

→ ESP32S3 Pro Microcontroller

Dual-core, Wi-Fi + BLE, low-power sleep modes

→ MPU6050 6-Axis IMU

Accelerometer + gyroscope for impact & lean-angle sensing

→ NEO-6M GPS Module

±2.5m accuracy, cold start <30s, 100Hz sampling

→ GSM SIM800L /LTE Uplink

SMS + data emergency alerts, works without smartphone

→ Li-Ion 2000mAh + IP67 Housing

Autonomous operation · all-weather rated

Future Hardware Enhancements

Edge Hardware

ESP8266 + MPU6050 (6-axis IMU) + NEO-6M GPS @100Hz sampling. ESP32-CAM on standby for visual evidence capture.

Cloud Stack

Node.js/Express hosted on Render. **MongoDB Atlas** for telemetry storage. Cloudinary CDN for crash footage delivery.

Block Chain

Block chain Services for the telemetry report on insurance company portal.

Calibration + Installation Steps

- Mount unit on frame
- Zero IMU baseline
- Set lean thresholds
- Run impact check
- Trigger test alert
- Verify cloud receipt

Pillars 01 & 02: Trip Report + Emergency Escalation

Tiered Emergency Escalation

1 **Tier 0 — 30s Local Override**

Rider can cancel alert if conscious and uninjured.

2 **Tier 1 — ≥30s: Family + Ambulance**

Automatic notification dispatched. Still cancellable if bike righted.

3 **Tier 2 — ≥30min: Police Dispatch**

Continuous GPS tracking active. Hospital lookup + SOS enabled.



Insurance Telemetry

- Rolling 30s buffer @10Hz — freezes on crash impact
- Auto-generates structured claims report
- Risk Score + fraud evidence

Safety Score Trends | Gamified Streaks | Route History | Idle Analytics

Pillars 03 & 04: Insurance + Guardian



Guardian Dashboard

- Consent-based live speed, location & trip history
- Geofence + speed threshold alerts
- Curfew alerts, theft alert, multi-guardian mode

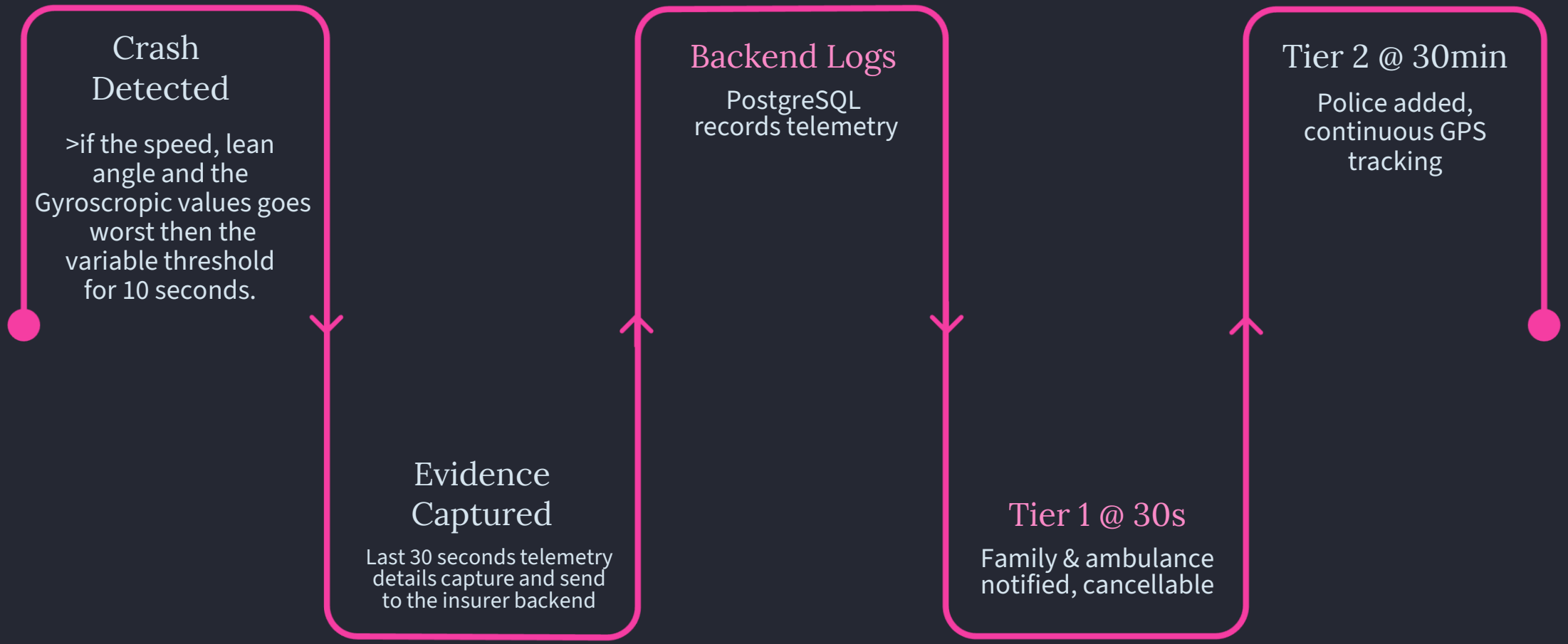


Trip & Journey Report

Per-trip logging: duration, distance, speed, harsh-braking events, idle time. Daily / weekly / monthly rollups.

CRASH LIFECYCLE

How a Crash Moves Through the System



Every second is engineered: from raw sensor threshold to multi-agency coordination, the system operates autonomously with zero manual intervention required.

Validated in Controlled Testing

92.6%

Detection Accuracy

Across 60 Simulated trials

71.1%

Detection Accuracy

Across 18 worst case trials

3

False Alerts

Zero false dispatches recorded

378ms

Mean Crash-to-DB Latency

Crash detected to MongoDB Atlas write

Validation Results — 60 Controlled Trials

Why NovaShields Plus Can Scale

Competitive Advantage Matrix

Feature	Smart Helmets	Phone Apps	Premium Bikes	NovaShields+
Auto Crash Detection	✗	✗	✓	✓
Live GPS Emergency Alert	✗	✓	✓	✓
Works Without Smartphone	✗	✗	✓	✓
Tamper-Proof Ride Logs	✗	✗	✗	✓
Affordable (<₹3,000)	✗	✓	✗	✓
Insurance-Ready Export	✗	✗	✗	✓
Retrofit Any Two-Wheeler	✗	✓	✗	✓
Tiered Escalation	✗	✗	✗	✓

Market Opportunity

Why Now?

- Two-wheelers = **75%** of India's motorized fleet
- GSM coverage >**95%** of highways
- IRDAI pushing telematics-based insurance



TAM

25 Cr+ two-wheelers registered in India

SAM

8 Cr connected-ready riders with GSM coverage

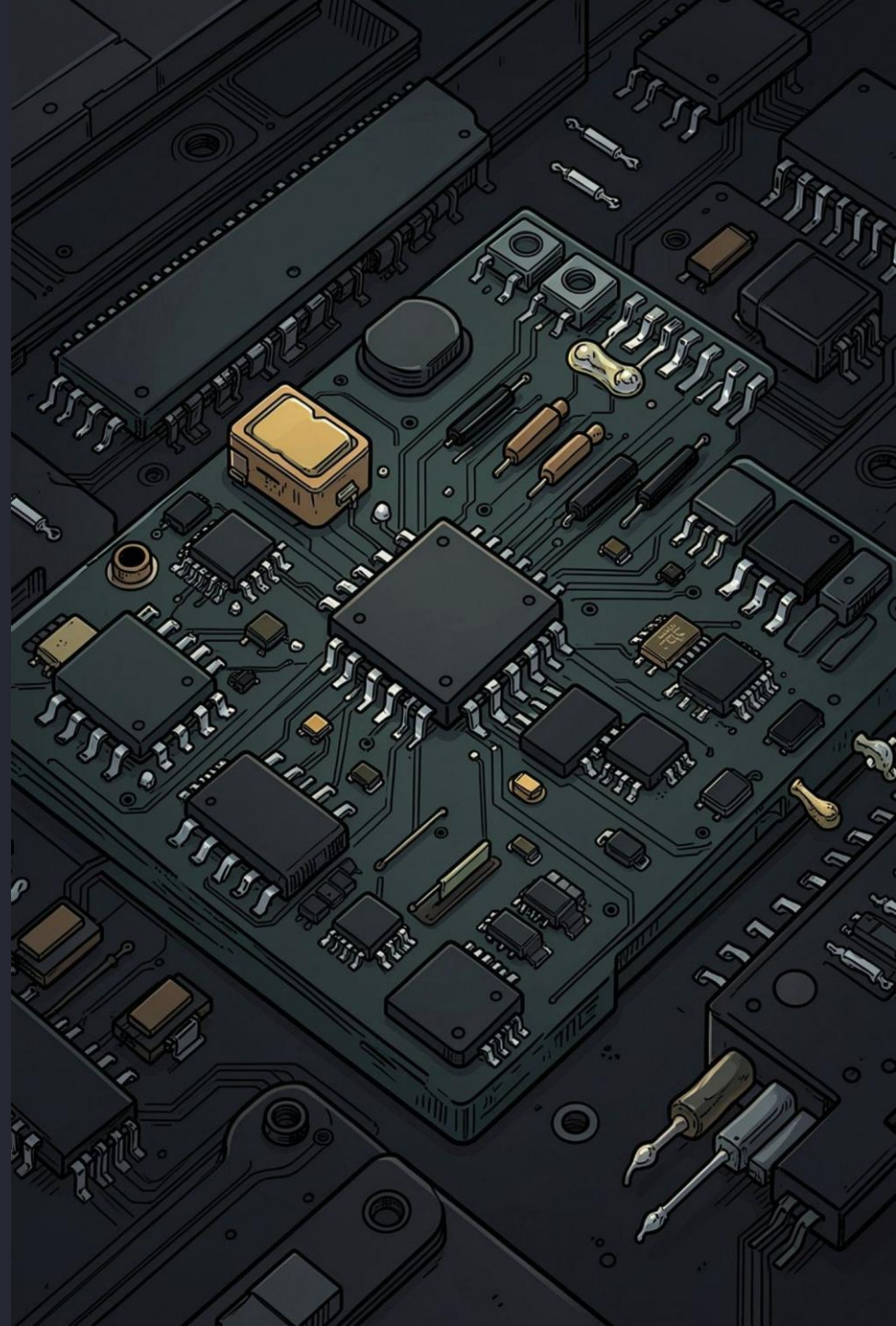
SOM

40 L target riders in years 1–3 (metros & tier-1 cities)

Bill of Materials — Prototype

A working prototype for under ₹3,000. Mass-production BOM projected below ₹1,500.

Component	Quantity	Unit Price	Total Price	Notes
Microcontroller board	1	₹420	₹420	Primary control unit
Temperature sensor	2	₹85	₹170	Environmental monitoring
Humidity sensor	1	₹120	₹120	Combined with temp sensing
Motion sensor	1	₹95	₹95	Detection and wake trigger
Resistor assortment	1	₹60	₹60	Mixed SMD values
Capacitor assortment	1	₹75	₹75	Decoupling and filtering
Prototype PCB	1	₹180	₹180	2-layer custom board
Connectors and headers	1	₹140	₹140	Board-to-wire interface
Voltage regulator	1	₹55	₹55	Stable power conversion
Passive support components	1	₹90	₹90	LEDs, diodes, ferrites
Assembly and test consumables	1	₹250	₹250	Solder, wire, standoffs
Total Prototype BOM	**13**	**₹1,570**	**₹1,575**	**Projected mass-mass-production below ₹1,500**



Revenue, Channels & Unit Economics

Three revenue streams. Retail today. OEM and insurance tomorrow.

$ REVENUE STREAMS

Device Sale

Rs. 2,499 MRR per device

Side Telematics Subscription

Rs. 99 / month + rider app

Insurer Data Licensing

Per-claim risk score API

CHANNELS

Dealership Retrofit

Bajaj, Hero, TVS Service Network

OEM Factory-Fit

White label pilot in year 2

Direct-to-Rider

Amazon, Flipkart — cost controlled

UNIT ECONOMICS

Rs. 2499 + Rs. 1188/yr = Rs. 3687

ARPU

46% @ 10K units

Gross Margin

12mo

Payback Period

Per rider

400K

Year 3 Target

Devices deployed

Year 3 target: 400,000 devices deployed • **$148 CR revenue run rate** • Break-even by month 22

Building Safer Roads for Every Rider

KEY TAKEAWAYS

Affordable Safety

Protection engineered for every motorcycle, at every price point.

Crash Detection & Alerts

Automatic incident detection with instant emergency notifications.

Rider Emergency Assistance

Real-time support features that respond when riders need help most.

Valuable Ride Data

Actionable insights for riders and insurers alike.

OUR VISION

> *"To make intelligent crash protection a standard feature on every two-wheeler — reducing injury and fatality through affordable technology."*



One Kit. Four Connected Safety Experiences. Zero Extra Sensors.

Thank you for your time.

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